

Unicode Pattern Search Engine for Indian Language Wikipedia

A Unicode-Aware Text Search Engine Using Data Structures and Algorithms

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ABSTRACT:

The **Unicode Pattern Search Engine** is designed to provide fast and efficient keyword searching across large **Indian-language Wikipedia datasets**. The system addresses the limitations of traditional substring searching, which can become inefficient when handling large multilingual corpora and complex Unicode scripts such as **Telugu, Devanagari, Tamil, and Bengali**. It uses **Unicode-aware text preprocessing** to handle multilingual content effectively.

The proposed system implements advanced **Data Structures and Algorithms**, including **KMP, Rabin-Karp, and Aho-Corasick** for efficient pattern matching. An **inverted index** is used to reduce search time and support large datasets, while **Hash Tables** assist in fast lookup and frequency-based processing. The system supports **single and multi-keyword queries**, ranks results based on relevance, and provides **contextual snippets** to help users quickly identify useful information.

Problem Statement

Indian language Wikipedia is expanding at an unprecedented pace, yet the tools to search it remain inadequate. Traditional substring search is too slow for large multilingual datasets, and Unicode scripts — Devanagari, Telugu, Tamil, Bengali — require specialized handling that most existing systems simply do not provide.



Core Challenge: Build a fast, scalable, Unicode-aware search engine using efficient DSA techniques.

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Rapid Content Growth

Indian language Wikipedia articles are growing rapidly, demanding scalable search infrastructure.

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Unicode Complexity

Scripts like Devanagari and Tamil require proper normalization before any search can be performed.

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Missing Capabilities

Existing systems lack efficient indexing and multi-keyword search support for Indian languages.



Key Algorithms and Data Structures

Pattern Matching Algorithms

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1

KMP

Single-pattern search with $O(n+m)$ time complexity. Avoids redundant comparisons using a failure function.

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Rabin-Karp

Hash-based multi-pattern detection. Efficient rolling hash enables simultaneous pattern scanning.

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3

Aho-Corasick

Multi-keyword matching in a single pass over the text. Ideal for large keyword sets.

Indexing Structures

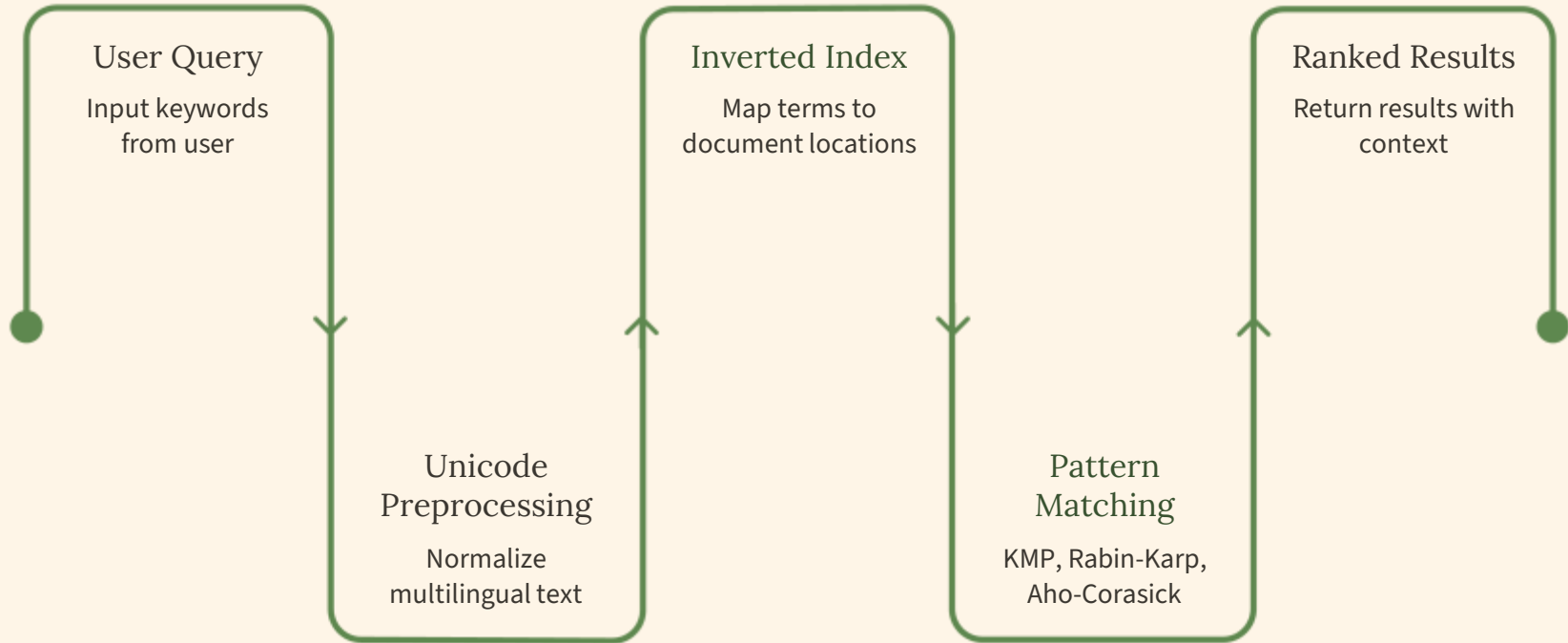
Inverted Index

Maps every keyword to the list of documents containing it, enabling sub-linear search time.

Hash Tables

Fast $O(1)$ average-case lookup and frequency counting across the document corpus.

Proposed system



Expected Outcomes & Future Scope

Expected Outcomes



Faster keyword searching in large multilingual collections



Unicode-aware support for multiple Indian languages



Efficient single and multi-keyword search



Relevant ranked results with contextual snippets

Future Scope



Advanced Ranking

Implement TF-IDF and BM25 scoring for more nuanced relevance ranking.



Web Interface

Build a web-based search interface accessible to end users across devices.



Distributed Processing

Scale to large datasets using distributed computing infrastructure.

CONCLUSION:

The **Unicode Pattern Search Engine** provides a powerful and scalable approach to searching large collections of **Indian-language Wikipedia content**. The project addresses the limitations of traditional substring searching by combining **Unicode-aware text processing with advanced Data Structures and Algorithms**.

By integrating **KMP, Rabin-Karp, and Aho-Corasick**, the system supports both **single and multi-keyword searching** efficiently. The **Inverted Index and Hash Tables** further reduce search time and make the system suitable for handling large multilingual datasets.

The system goes beyond simple keyword matching by providing **relevance-based ranking and contextual snippets**, helping users quickly identify the most useful information. Its support for **Telugu, Devanagari, Tamil, Bengali, and other Unicode scripts** makes it particularly relevant to multilingual information retrieval.